Khalil LAGHA

M2 student in Electrical Energy Systems Design — power electronics & energy conversion · Université Grenoble Alpes

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PROFILE

M2 student in Electrical Energy Systems Design (CSEE, UGA), **ranked 2nd of the M1 EEA cohort (15.43/20)**. Specialized in **power electronics**: **DC-DC converters (buck, boost)**, AC-DC rectification, **regulation-loop tuning (PI/PID)**, **efficiency optimization**, EMI compliance — validated in simulation (Matlab/Simulink) then on real hardware, across **five internships** from the lab (GIPSA-lab) to industrial sites. Target fields: **EV / powertrain**, **electrical grids**, **AI applied to power systems**. **Seeking: work-study contract (alternance) from September 2026 — 2–3 week company / university rotation.**

WORK EXPERIENCE

Research Intern — *GIPSA-lab, Grenoble* — 04–07/2026 · 3.5 months

- Modeled a **PEMFC fuel cell** in Matlab/Simulink — **6 %** dynamic and **1 %** static error against real measurements.
- Tuned the **regulation loop** (PI controller) of a **DC-DC boost converter** — regulated voltage, optimized conversion efficiency; validated in simulation then on a real test bench.

Intern — *SLB / Schlumberger* — 07/2025 · 1 month

- Authored technical summaries **in English** (directional drilling) in an international industrial environment; earned the **NEST & SIPP Level 2** HSE certifications.

Power Electronics & Automation Intern — *EURL Lagha* — 01/2024 · 15 days + 09/2024 · 1 month

- Contributed to an **industrial DC-DC buck converter** — 50 V → 0–48 V adjustable, **25 A**.
- Worked on **4 transformer-rectifiers** feeding pipelines (electrolysis) — three-phase 400 V → 100 V transformer, rectification + capacitive filtering: **100 V DC / 80 A (≈ 8 kW)** output, ripple < 5 %, **EMI compliance**.
- Co-designed the supervision HMI (**Siemens TIA Portal**, Ladder PLC); produced the electrical schematics in **EPLAN**; studied a transformer against a customer specification.

Power Grid Intern — *Sonelgaz* — 03/2024 · 15 days · observation

- Studied the full generation → distribution chain (**400/220 kV → 60 kV → 30/10 kV → 400/230 V**) and substation architecture; visited an MV/LV substation.

EDUCATION

M2 CSEE — Electrical Energy Systems Design · *Université Grenoble Alpes* — 09/2026 · admitted · work-study

M1 EEA (EECS) · *Université Grenoble Alpes* — 2025 – 2026

Ranked 2nd of the cohort · annual average **15.43/20**.

Engineering programme in Electrical Engineering · *École Nationale Polytechnique, Algiers* — 2021 – 2025

4 of 5 years completed · moved to France for the master's · prep classes: among the top-ranked.

Baccalauréat in Mathematics — 2021

Highest honours (Très Bien) — **16.93/20**.

PROJECTS

Power grid analysis — Matlab · solo FLAGSHIP

- Developed **load flows from 3 to 118 buses** (PQ/PV/slack; IEEE 14/30/57/118 networks); compared **3 numerical methods** — Gauss-Seidel, Newton-Raphson, fast-decoupled (accuracy vs computation time); studied **transient stability** (critical fault-clearing time).

Field-oriented control (FOC) of an induction motor — Matlab/Simulink

- Implemented Park transforms, PID controllers and observers; tuned the **cascaded current- and speed-regulation loops**.

Electrical machine sizing — Python · team

- Built a **Python (OOP, PyQt)** application — stator winding design, flux density via **reluctance networks**, finite-element validation (**FEMM**), automatic parameter generation.

Neural network from scratch (MLP) — Matlab · open source

- Hand-coded a multilayer perceptron and its **backpropagation** without any toolbox; debugged, tested and **published under the MIT license** — github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab.

Also: cascade control of a DC motor — three-phase thyristor bridge, outer speed / inner current loop (Simulink); Siemens Step7 automation project (HMI + Ladder); Arduino line-follower robot (Poly Maze competition).

TECHNICAL SKILLS

Power electronics: DC-DC conversion (buck, boost), AC-DC (diode and thyristor rectification), DC-AC, AC-AC · converter sizing · PWM · filtering, ripple · EMI compliance · efficiency optimization

Control engineering: regulation-loop tuning (PI/PID), LQR, observers, Kalman filter · field-oriented control (FOC) · cascade control · system identification

Machines & grids: electrical machines and transformers (analysis, design) · power system analysis — load flow, Y/Z matrices, transient stability · MV/LV distribution

Industrial automation: Siemens PLC (TIA Portal, Step7 — Ladder, SCL, FBD, SFC) · HMI/SCADA · Grafcet

AI: neural networks (backpropagation), genetic algorithms, fuzzy logic

SOFTWARE

Matlab/Simulink (Simscape Power Systems) · Python (OOP, PyQt) · C/C++ · LTspice · Proteus · EPLAN · AutoCAD Electrical · Siemens TIA Portal · Step7 (Ladder/SCL) · FEMM · SolidWorks · Arduino · Git/GitHub

SOFT SKILLS & ACTIVITIES

Fast learner · comfortable with new tools & software · problem solving · rigorous · organized — IEEE Student Branch ENP · VIC (ENP)

LANGUAGES

French — C1 · English — C2 · Arabic — native

CERTIFICATIONS

SLB — NEST & SIPP Level 2 (07/2025) — HSE, industrial operations safety